

Nouredine Khaous

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PhD Student in Computational Linguistics specializing in speech and language representation learning. My research investigates how neural models encode linguistic information through the analysis of internal representations, and develops evaluation methods that better capture linguistic structure in multilingual and low-resource settings.

WORK EXPERIENCE

- Data Scientist** Sep 2023 – Sep 2025
Orange  Paris, France
 - Adapted large language models to the Wolof language for low-resource NLP applications by building datasets, performing vocabulary expansion, and supervising multi-GPU fine-tuning with DeepSpeed.
 - Achieved a +12% METEOR improvement on Wolof→French and Wolof→English machine translation tasks.
 - Built and deployed an Agentic RAG system with LangGraph for network incident ticket management, integrating summarization, chatbot support, and next-action prediction.
 - Reduced customer-care ticket processing time by 50% and automated invoice processing workflows, reducing handling time by 75%.
- Research Intern** May 2025 – Jul 2025
Bergische Universität Wuppertal Remote
 - Collaborating on AI models integrating physics for fluid state prediction in nonlinear port-Hamiltonian systems.
 - Developing conditional diffusion models to generate spatially-resolved system state predictions in image form.
 - Supporting system identification and simulation of complex fluid dynamics processes.
 - Conducting research at the intersection of machine learning, physics-informed AI, and dynamical systems.
- Research Intern** Oct 2024 – Jan 2025
IDEMIA  Paris, France
 - Improved contactless fingerprint image quality using state-of-the-art diffusion models for biometric applications.
 - Designed an attention-based U-Net architecture with time-step encoding for image denoising.
 - Enhanced similarity between contactless and contact-based fingerprint images, improving recognition robustness.
 - Conducted experiments on generative image restoration methods for secure identity systems.

EDUCATION

- PhD in Machine Learning for Low-Resource Languages** Dec 2025 – Present
Université Paris Cité | Laboratoire de Linguistique Formelle Paris, France
 - Topic: Machine learning for rare languages: linking speech recognition and analysis of the internal representations of Speech Models in service of field linguistics | Supervisor: Guillaume Wisniewski
- Master of Science in Computer Science and Robotics** Sep 2023 – Sep 2025
Sorbonne Université Paris, France
 - Focus on Machine Learning, Speech Processing, Computer Vision and Robotics.
 - Dissertation Topic: Adapting Large Language Models to Low-Resource Languages: A Case Study of Wolof.
 - Graduated Rank: 3/45
- Bachelor of Engineering in Electronics and Control Theory** Sep 2021 – May 2023
Université Claude Bernard Lyon 1 Lyon, France
 - Focus on Electronics, Electrical Engineering, and Control Theory.
 - Graduated Rank: 4/50
- Preparatory Classes for Engineering Schools | Mathematics and Physics Track** Sep 2019 – Jun 2021
École Nationale Polytechnique Algiers, Algeria
 - Intensive coursework in Mathematics, Linear Algebra, Physics, Chemistry, and Engineering Sciences.
 - Graduated Rank: 19/283

SKILLS

- Programming Languages:** Python, C/C++, SQL, Bash
- Machine Learning & AI:** PyTorch, Hugging Face, Deep Learning, NLP, LLMs, Transformers, Computer Vision
- Data Science & Scientific Computing:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- Tools & Technologies:** Docker, Git, Linux, OpenCV, LaTeX
- Research Areas:** Speech Processing, Representation Learning, Low-Resource Languages, Multilingual NLP
- Languages:** French (Native), Arabic (Native), Kabyle (Native), English (Full Professional Proficiency)

PUBLICATIONS

- **Khaous, N. & Wisniewski, G.** (2025). *Accent-Induced Geometric Displacement in Speech Representations Predicts Automatic Speech Recognition Degradation*. TALN 2025.